AskDocs AI

PROJECT PHASE II

MASTER OF COMPUTER APPLICATIONS

(MCA -IV SEM)

SUBMITTED BY

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Enrolment No.-U2449028

Batch Year-2024-26

Under the Supervision of,

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CERTIFICATE

This is to certify that **Kshitij Gupta**, student of the **Master of Computer Applications (Semester –IV)** of the Centre of Computer Education & training, IPS, UoA, has completed his work in the project entitled **"AskDocs AI**" under my guidance in the academic session (2025–26).

He has taken proper care and shown utmost sincerity in the completion of this project. I certify that his work on this project is up to my expectations and as per the guidelines. This project is original and is never submitted somewhere else for the same purpose.

DATE: **MS. KOMAL SINGH**

Place: CCET, IPS, UoA, Prayagraj (Project Supervisor)

SELF -DECLARATION

I hereby declare that the project entitled "**AskDocs AI"** is my original work and has not been submitted, in part or in full, for any other degree. All the information and data presented in the report are authentic and have been surely acknowledged.

I confirm that all the sources consulted during my work have been duly acknowledged in the reference section, and the report does not contain any plagiarized content. I understand that any act of plagiarism, including copying or paraphrasing the work of others without proper citation, is considered academic misconduct and can lead to severe penalties. I also understand that any misrepresentation or falsification of information can lead to the rejection of my project report and can affect my academic standing.

DATE: **KSHITIJ GUPTA**

Place: CCET, IPS, UoA, Prayagraj (Enrollment No.-U2449028)

ACKNOWLEDGEMENT

I would like to take this opportunity to express my gratitude to everyone who has supported us in completing this project.

Firstly, I would like to express my sincere gratitude to **Dr. Nilesh Anand Srivastava,** the coordinator of the **Centre of Computer Education** at the **University of Allahabad**, for his invaluable guidance and support throughout my project work.

Foremost, I would like to thank our project supervisor, **Ms. Komal Singh**, for her invaluable guidance, feedback, and encouragement throughout the project. I would also like to thank subject teachers **Ms. Sunita Tripathi** who has constantly resolved our domain-specific queries throughout the project. Their expertise and experience have been instrumental in shaping our understanding of the subject and developing our research skills.

I am also grateful to my family and specially my friends for their constant support and motivation, which has kept me going throughout the project. Their belief in me and my abilities has been a source of inspiration and encouragement.

PROJECT REPORT

1. INTRODUCTION

Project Title: AskDocs AI

Category: AI-Powered Web Application (Semantic Q&A System)

With the exponential growth of digital information, the ability to quickly extract actionable insights from documents is paramount. Students struggle to find specific concepts across semesters of notes; researchers spend hours re-reading papers to locate a single statistic; legal professionals manually sift through case files. AskDocs AI is conceived to address this "information overload" by building a bridge between unstructured document storage and intuitive, conversational knowledge retrieval.

AskDocs AI is a web-based intelligent document question-answering system that allows users to upload documents such as PDFs, text files, and reports, and then ask questions in natural language to retrieve precise, context-aware answers. Unlike traditional search tools (Ctrl+F) which rely on exact lexical matching, AskDocs AI utilizes semant. It understands the intent and meaning behind the query.

1.1 PROBLEM DEFINITION

In today's digital world, organizations and individuals handle a large volume of documents such as PDFs, reports, manuals, and knowledge bases. Extracting useful information from these documents is often time-consuming, inefficient, and requires manual effort.

Traditional search methods rely on keyword matching, which:

1. Do not understand the context or meaning of the query
2. Return irrelevant or excessive results
3. Require users to read entire documents to find answers

Moreover, non-technical users face difficulty in navigating complex documents and retrieving precise information quickly.

There is a need for a system that can:

1. Understand natural language queries
2. Provide accurate and context-aware answers
3. Reduce the time spent in searching and reading documents
4. Work efficiently across multiple document formats

1.2 MOTIVATION

In the modern digital era, a vast amount of information is stored in the form of documents such as PDFs, reports, manuals, and knowledge bases. However, accessing relevant information from these documents remains a major challenge. Users often spend significant time manually searching, reading, and extracting useful content, which reduces productivity and efficiency.

The motivation behind developing AskDocs AI arises from the need to simplify and automate this process. Traditional search systems are limited to keyword-based retrieval and fail to understand the actual intent of the user's query. This often leads to irrelevant results and user frustration.

With the rapid advancement in Artificial Intelligence (AI) and Natural Language Processing (NLP), it is now possible to build intelligent systems that can understand human language and provide accurate, context-aware responses. This creates an opportunity to transform the way users interact with documents.

The key motivations for this project include:

1. To reduce the time and effort required to search through large documents
2. To provide intelligent, context-based answers instead of simple keyword matches
3. To enhance user experience by enabling natural language interaction
4. To make information retrieval accessible even for non-technical users
5. To leverage modern AI technologies for practical, real-world problem solving

Thus, AskDocs AI is motivated by the goal of creating a smart, efficient, and user-friendly document question-answering system that improves productivity and decision-making.

2. OBJECTIVE

To develop an AI-powered system, AskDocs AI, that enables users to retrieve accurate, context-aware information from documents using natural language queries.

3. REQUIREMENT ANALYSIS AND SPECIFICATION

3.1 Functional Requirement:

1. **USER REGISTRATION & LOGIN**
   The system shall allow users to register and log in securely using their email and password.
2. **USER AUTHENTICATION**
   The system shall authenticate users and provide access using JWT tokens.
3. **DOCUMENT UPLOAD**
   The system shall allow authenticated users to upload documents in PDF and TXT formats.
4. **DOCUMENT STORAGE**
   The system shall store uploaded documents along with their metadata in the database.
5. **TEXT EXTRACTION & PROCESSING**
   The system shall automatically extract and process text from uploaded documents.
6. **EMBEDDING GENERATION**
   The system shall convert processed text into vector embeddings for efficient search.
7. **QUERY HANDLING (Q&A INTERFACE)**
   The system shall allow users to ask questions in natural language through an API.
8. **ANSWER RETRIEVAL & RESPONSE**
   The system shall retrieve relevant information and return accurate answers along with source details.

3.2 Non-Functional Requirement:

1. **PERFORMANCE**
   The system shall provide fast response times, with the Q&A module responding within a few seconds under normal load conditions.
2. **SCALABILITY**
   The system shall be capable of handling increasing numbers of users and documents without significant performance degradation.
3. **SECURITY**
   The system shall ensure secure authentication using JWT tokens and protect user data through encryption and password hashing.
4. **USABILITY**
   The system shall provide a user-friendly interface and clear API responses for easy interaction by both technical and non-technical users.
5. **RELIABILITY**
   The system shall ensure consistent operation with minimal failures and proper error handling mechanisms.
6. **MAINTAINABILITY**
   The system shall be modular and well-structured to allow easy updates, debugging, and future enhancements.
7. **AVAILABILITY**
   The system shall always be available to users with minimal downtime.
8. **DATA INTEGRITY & PERSISTENCE**
   The system shall ensure that all documents, embeddings, and user data are stored reliably and remain consistent across system operations.

Technical Requirements:

Category	Requirement
Operating System	Windows 10/11, macOS, or Ubuntu 20.04+
Programming Language	Python 3.10 or above
Web Framework	Django 4.2
API Framework	Django REST Framework (DRF)
Database	PostgreSQL
Vector Database	FAISS (Facebook AI Similarity Search)
NLP Library	Sentence-Transformers
PDF Processing Library	Pdfplumber
Authentication	JWT (djangorestframework-simplejwt)
Version Control	Git & GitHub
IDE/Editor	Visual Studio Code
Containerization	Docker & Docker Compose

3.3 Software & Hardware Requirements:

1. Hardware Required:

Category	**Minimum Requirement**
Processor	Intel i3 (10th Gen) / AMD equivalent
RAM	4 GB
Storage	8 GB free space
Internet	5 Mbps connection
Category	Minimum Requirement
Processor	Intel i3 (10th Gen) / AMD equivalent

2. Software Required:

Category	**Specification**
Operating System	Win Windows 10/11, macOS, or Ubuntu Linux
Programming Language	Python 3.10+
Web Frameworks	Django 4.x + Django REST Framework
Database	PostgreSQL 14+
Authentication	Django rest framework- simple jwt (JWT)
Containerization	Docker & Docker Compose
IDE	VS Code
Version Control	Git + GitHub

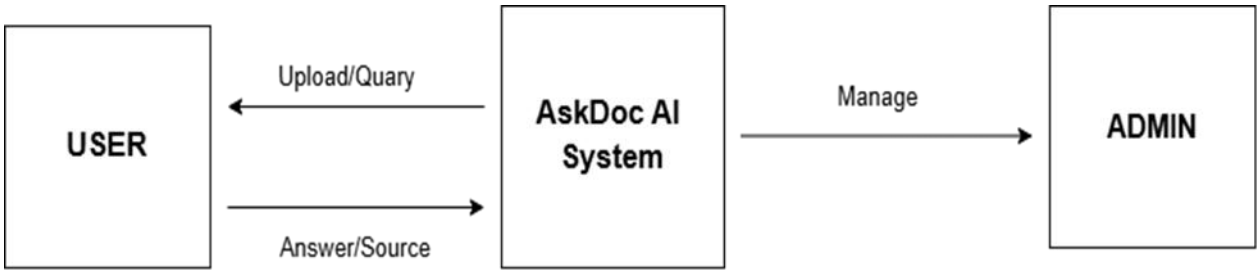
3.4 CODE (MODULE-WISE)

Module	Time (Weeks)	Description
Planning and Design	1-2	Define project scope, objectives, and requirements. Design overall system architecture for document processing and Q&A pipeline.
Environment Setup	1	Install and configure Python, Django, PostgreSQL, FAISS, and required NLP libraries. Set up virtual environment and project structure.
User Management Module	1-2	Implement user registration, login, JWT authentication, and role-based access control (Admin/User).
Document Management Module	2	Develop document upload, storage, metadata handling, and document listing & deletion features.
Embedding & Indexing Module	2	Generate vector embeddings using Sentence-Transformers and store them in FAISS index along with mapping in database.
Q&A Engine Module	2-3	Implement question processing, vector search (k-NN), and retrieval of relevant document chunks.
API & Interface Integration	1	Develop REST APIs and integrate all modules to ensure smooth communication between components.

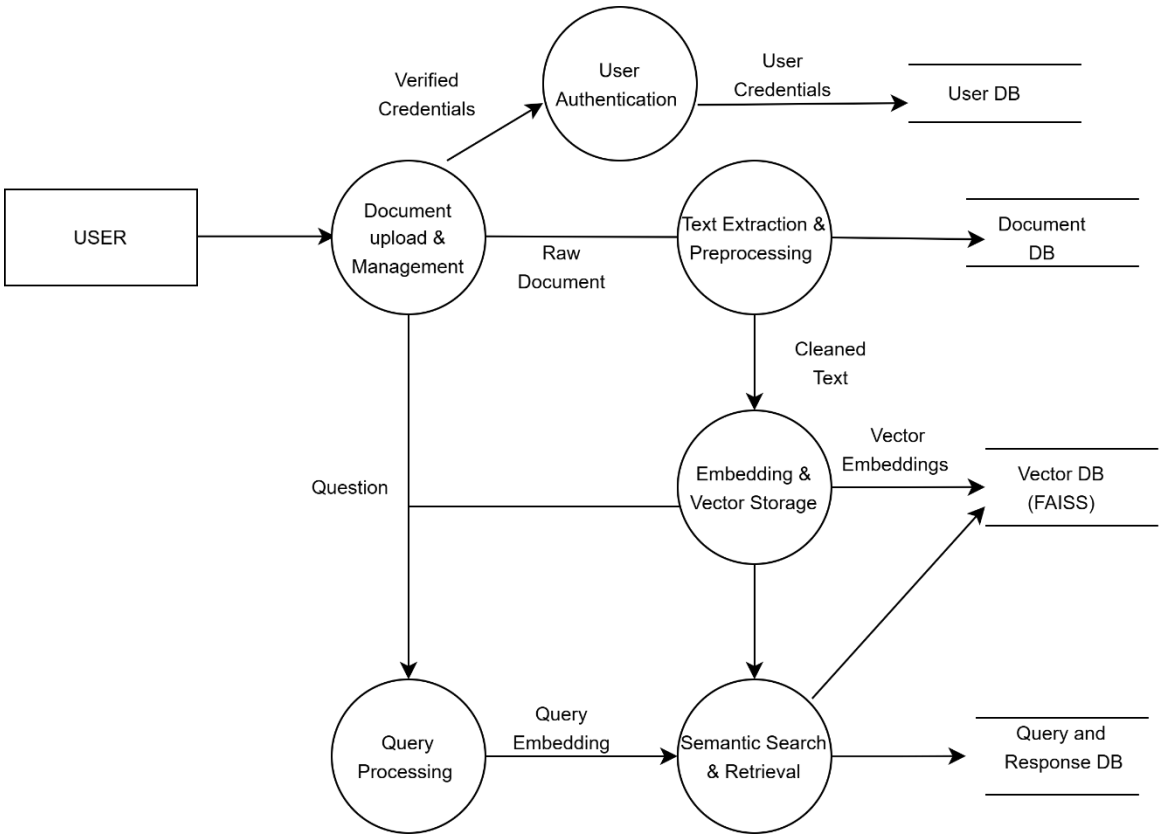
4. SYSTEM DESIGN

4.1 DATA FLOW DIAGRAM

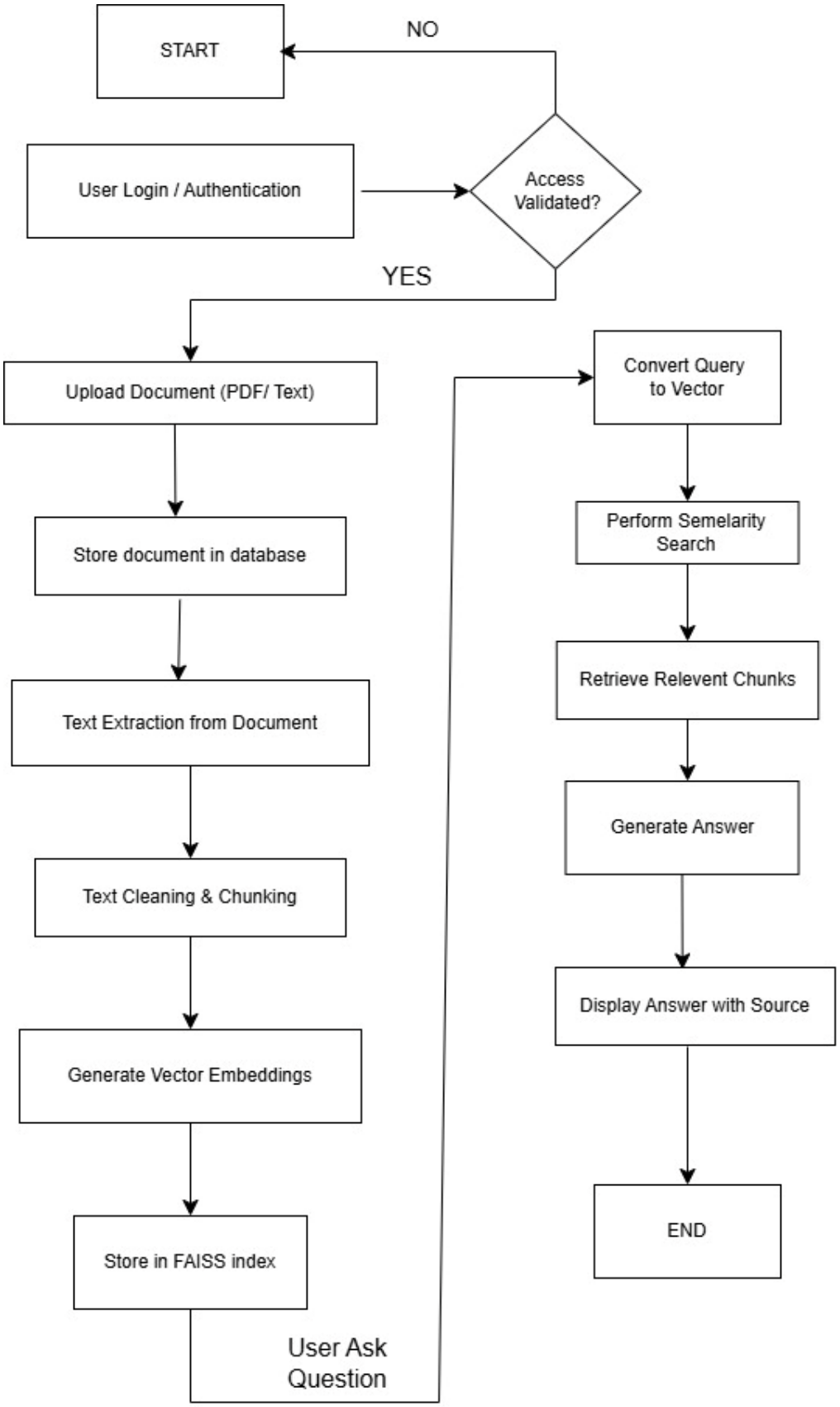
LEVEL 0 DFD



LEVEL 1 DFD



4.2 Flowchart



6. DEVELOPMENT & TESTING

Testing Strategy	Area	Details
1. Unit Testing	Backend APIs	User registration, login, JWT authentication
	Document Module	File upload, storage, metadata handling
	Processing Module	Text extraction, cleaning, chunking
	Embedding Module	Vector generation using Sentence-Transformers
	Q&A Module	Query processing and FAISS search
2. Integration Testing	Module Interaction	Authentication then Document Upload then Processing
		Processing then Embedding then FAISS Index
		Query then Vectorization then Similarity Search
		Backend then PostgreSQL then FAISS
3. Functional Testing	User Features	Register, login, upload documents
		Ask questions and get answers
		View and delete documents
	System Output	Accurate answers with source citation
4. API / Interface Testing	API Endpoints	Correct JSON responses and status codes
		Authentication validation for protected routes

	Error Handling	Invalid file type, missing query, unauthorized access
5. Usability Testing	User Experience	Easy document upload and query process
		Clear and understandable responses
		Minimal technical knowledge required
6. Performance Testing	Response Time	Q&A response within 3 seconds
	System Load	Handles multiple users simultaneously
	Efficiency	Fast similarity search using FAISS
7. Edge Case Testing	Input Cases	Empty query, large files, unsupported formats
	System Cases	Missing FAISS index, no relevant results
	Security Cases	Expired or invalid JWT tokens

7.CODE

1.ASKDOCS AI/static/frontend/index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>AskDocs AI</title>
    <link rel="stylesheet" href="/css/style.css">
    <link rel="stylesheet" href="/css/app.css">
</head>
<body>

<!-- Auth Guard: redirect to login if not logged in -->
<div id="app" style="display:none; width:100%; height:100vh;">

    <!-- ── SIDEBAR ──────────────────────────────────────────────── -->
    <aside class="sidebar" id="sidebar">

        <!-- Logo -->
        <div class="sidebar-logo">
            <span class="logo-icon"></span>
            <span class="logo-text">AskDocs AI</span>
        </div>

        <!-- Upload Button -->
        <button class="upload-btn" onclick="openUploadModal()">
            <span>+</span> Upload Document
        </button>
```

```
    <!-- Documents List -->
    <div class="sidebar-section">
        <div class="sidebar-section-title"> Documents</div>
        <div class="doc-list" id="docList">
            <div class="empty-state-small">No documents yet</div>
        </div>
    </div>

    <!-- Chat History -->
    <div class="sidebar-section">
        <div class="sidebar-section-title"> Recent Questions</div>
        <div class="history-list" id="historyList">
            <div class="empty-state-small">No history yet</div>
        </div>
    </div>

    <!-- Sidebar Footer -->
    <div class="sidebar-footer">
        <div class="user-info">
            <div class="user-avatar" id="userAvatar">K</div>
            <span class="username" id="usernameDisplay">User</span>
        </div>
        <button class="logout-btn" onclick="handleLogout()">
            Logout
        </button>
    </div>

</aside>

<!-- ── MAIN CONTENT ──────────────────────────────────────── -->
```

```
<main class="main-content">

    <!-- Top Bar -->
    <div class="topbar">
        <button class="menu-toggle" onclick="toggleSidebar()">☰</button>
        <h2 class="topbar-title">Chat with your Documents</h2>
        <div class="topbar-right">
            <span class="status-dot"></span>
            <span class="status-text">AI Ready</span>
        </div>
    </div>

    <!-- Chat Area -->
    <div class="chat-area" id="chatArea">

        <!-- Welcome Message -->
        <div class="welcome-screen" id="welcomeScreen">
            <div class="welcome-icon"></div>
            <h2>Welcome to AskDocs AI</h2>
            <p>Upload a PDF document and start asking questions.<br>
            The AI will find relevant answers with source references.</p>
            <div class="welcome-tips">
                <div class="tip"> Upload a PDF from the sidebar</div>
                <div class="tip"> Ask any question about your document</div>
                <div class="tip"> Get answers with page references</div>
            </div>
        </div>

        <!-- Messages will appear here -->
        <div class="messages-container" id="messagesContainer"></div>
```

```
</div>

<!-- Input Area -->
<div class="input-area">
  <div class="input-wrapper">
    <textarea
      id="questionInput"
      placeholder="Ask a question about your documents..."
      rows="1"
      onkeydown="handleKeyDown(event)"
      oninput="autoResize(this)"
    ></textarea>
    <button
      class="send-btn"
      id="sendBtn"
      onclick="sendQuestion()"
    >
      <svg width="20" height="20" viewBox="0 0 24 24" fill="none">
        <path d="M22 2L11 13" stroke="white" stroke-width="2"
            stroke-linecap="round" stroke-linejoin="round"/>
        <path d="M22 2L15 22L11 13L2 9L22 2Z" stroke="white"
            stroke-width="2" stroke-linecap="round"
            stroke-linejoin="round"/>
      </svg>
    </button>
  </div>
  <div class="input-hint">Press Enter to send · Shift+Enter for new line</div>
</div>
```

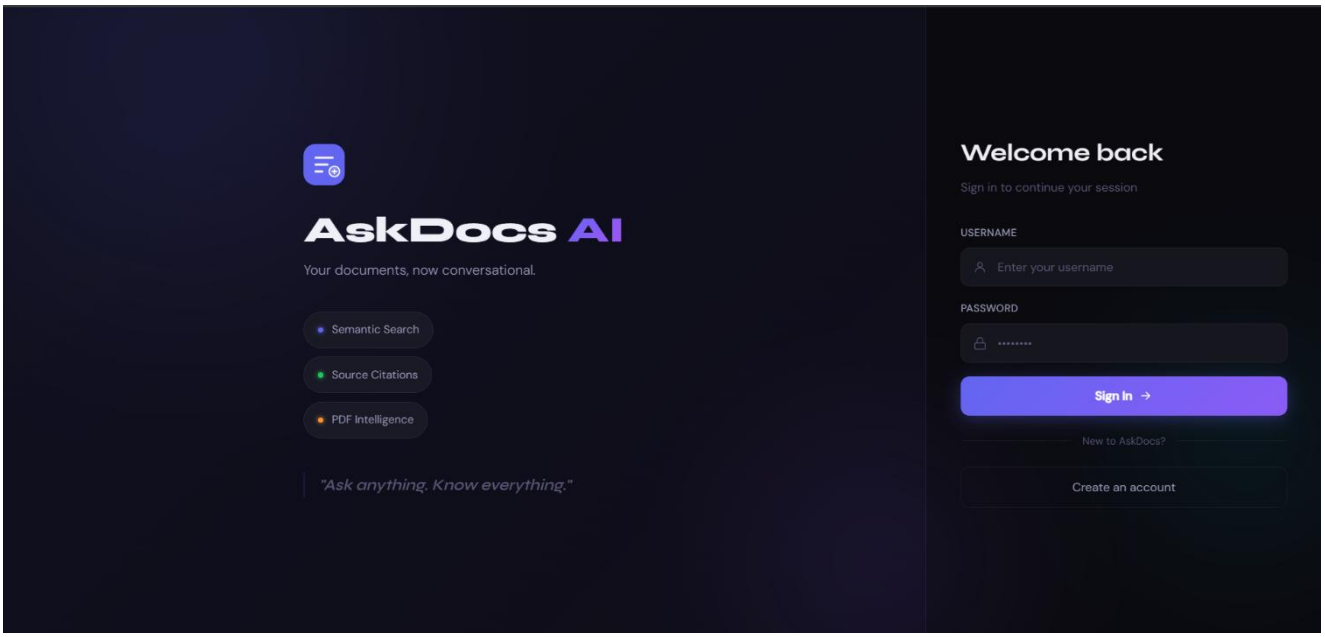
```
    </main>

</div>

<!-- ── UPLOAD MODAL ──────────────────────────────────────────── -->
<div class="modal-overlay" id="uploadModal">
    <div class="modal">
        <div class="modal-header">
            <h3>Upload Document</h3>
            <button class="modal-close" onclick="closeUploadModal()">×</button>
        </div>

        <div class="modal-body">
            <!-- Drop Zone -->
            <div class="drop-zone" id="dropZone" onclick="triggerFileInput()">
                <div class="drop-icon"></div>
                <p class="drop-text">Click to choose a PDF file</p>
                <p class="drop-hint">Supported format: PDF only · Max 10MB</p>
                <input type="file" id="fileInput" accept=".pdf"
                       style="display:none" onchange="handleFileSelect(event)"/>
            </div>

            <!-- Selected File Info -->
            <div class="file-info" id="fileInfo" style="display:none">
                <span class="file-icon"></span>
                <div class="file-details">
                    <span class="file-name" id="fileName"></span>
                    <span class="file-size" id="fileSize"></span>
                </div>
                <button class="file-remove" onclick="removeFile()">×</button>
```

```
</div>

<!-- Title Input -->
<div class="form-group" id="titleGroup" style="display:none">
  <label>Document Title (optional)</label>
  <input type="text" id="docTitle"
       placeholder="Leave blank to use filename"/>
</div>

<!-- Progress -->
<div class="upload-progress" id="uploadProgress" style="display:none">
  <div class="progress-step" id="step1">
    <span class="step-icon"></span>
    <span>Uploading file...</span>
  </div>
  <div class="progress-step" id="step2">
    <span class="step-icon"></span>
    <span>Extracting text...</span>
  </div>
  <div class="progress-step" id="step3">
    <span class="step-icon"></span>
    <span>Generating embeddings...</span>
  </div>
  <div class="progress-step" id="step4">
    <span class="step-icon"></span>
    <span>Building search index...</span>
  </div>
</div>

<!-- Error/Success in modal -->
```

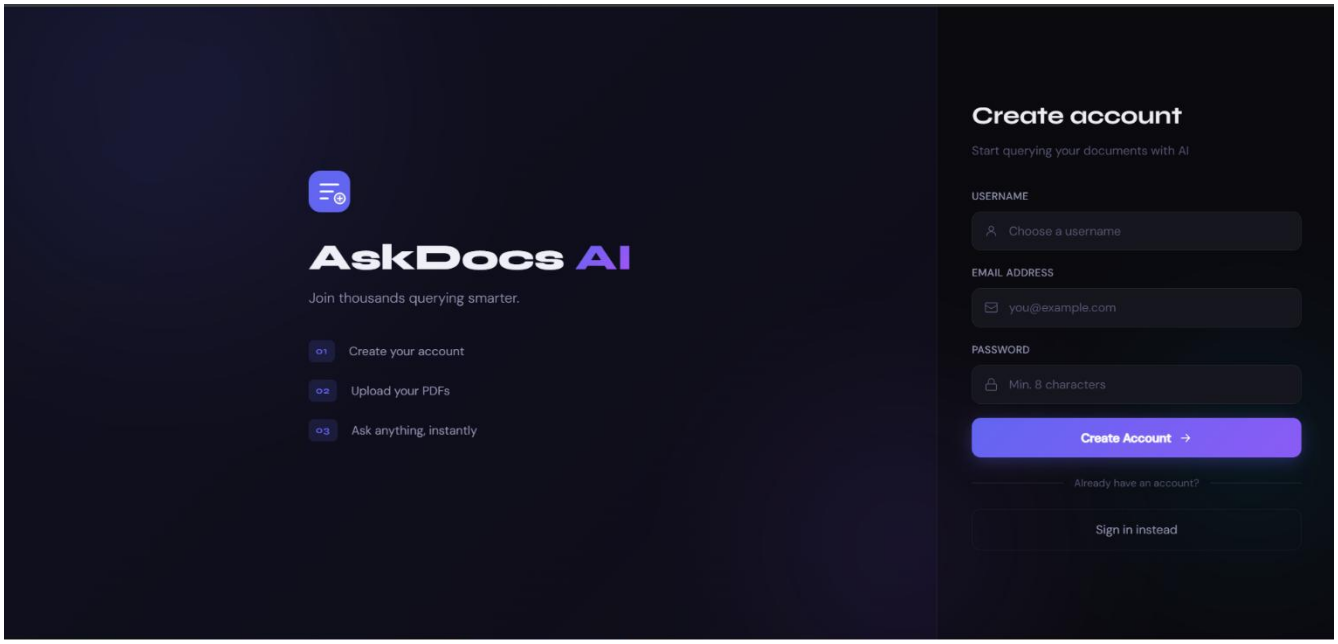
```
            <div class="error-msg" id="uploadError"></div>
            <div class="success-msg" id="uploadSuccess"></div>
        </div>

        <div class="modal-footer">
            <button class="btn-secondary" onclick="closeUploadModal()">
                Cancel
            </button>
            <button class="btn-primary" id="uploadBtn"
                    onclick="uploadDocument()" style="width:auto; padding: 12px 28px;">
                Upload
            </button>
        </div>
    </div>
</div>

<script src="/js/auth.js"></script>
<script src="/js/app.js"></script>

</body>
</html>
```

8. SNAPSHOTS

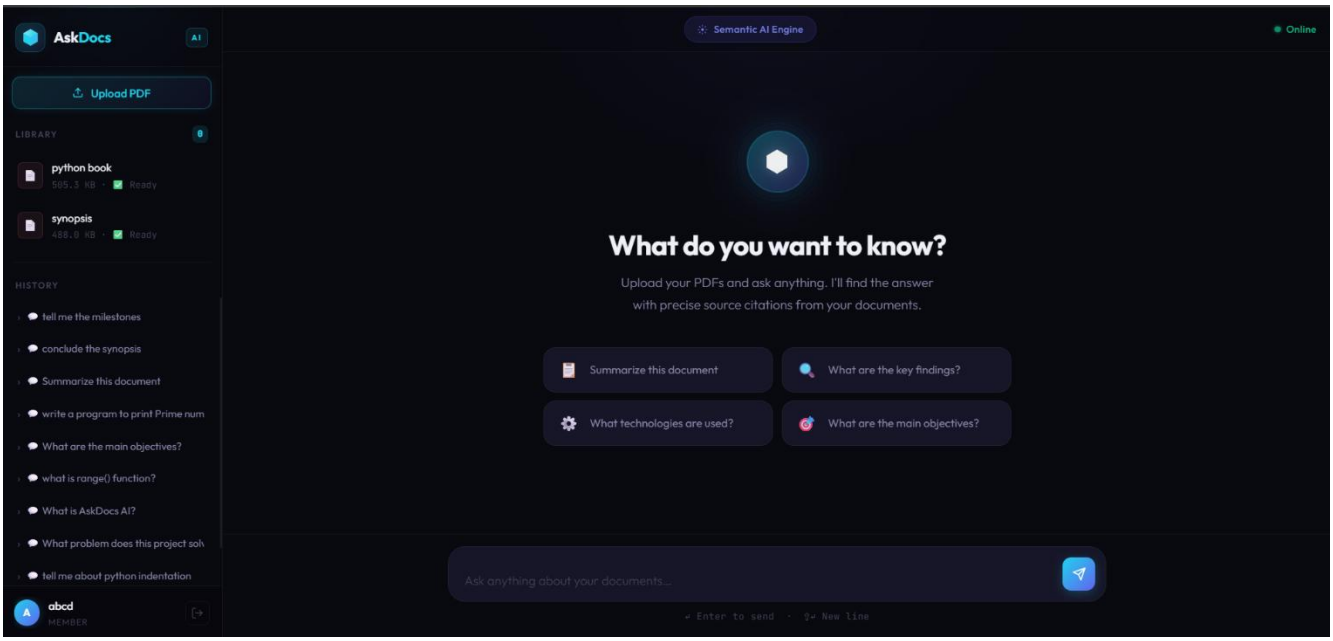
Login Page:



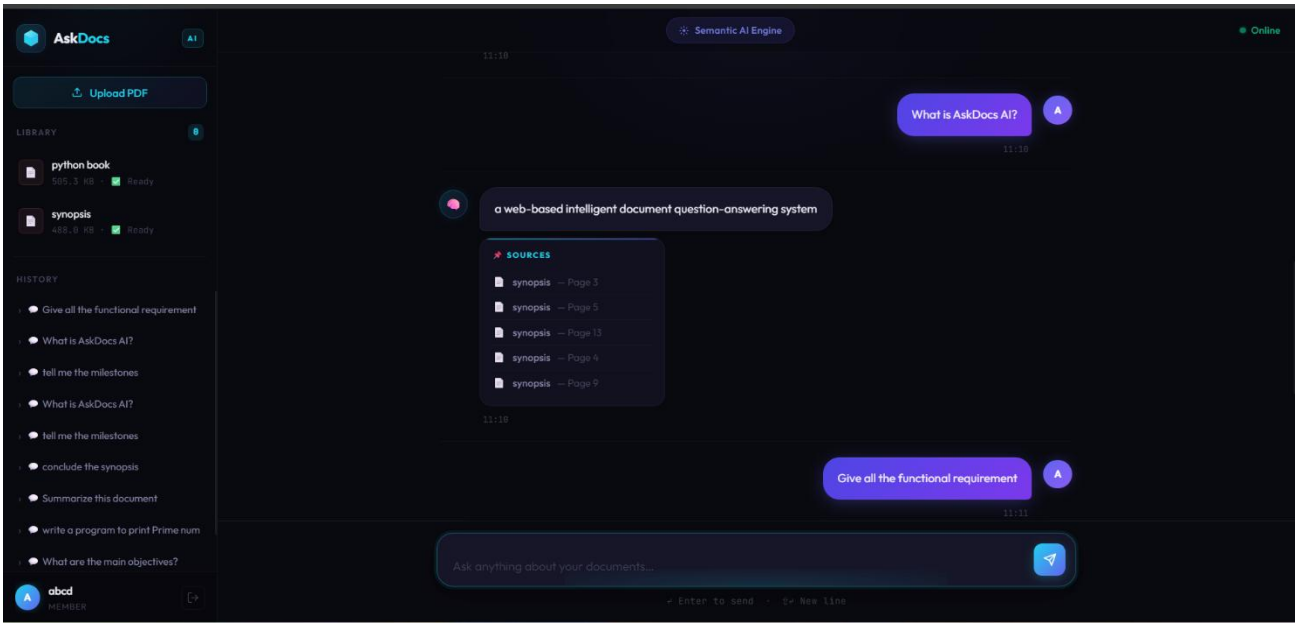
Register Page:



Home Page:


AskDocs
AI
Semantic AI Engine
Online
Upload PDF
LIBRARY
0
python book
505.3 KB · Ready
synopsis
488.0 KB · Ready
HISTORY
tell me the milestones
conclude the synopsis
Summarize this document
write a program to print Prime num
What are the main objectives?
what is range() function?
What is AskDocs AI?
What problem does this project solv
tell me about python indentation
abcd
MEMBER
What do you want to know?
Upload your PDFs and ask anything. I'll find the answer
with precise source citations from your documents.
Summarize this document
What are the key findings?
What technologies are used?
What are the main objectives?
Ask anything about your documents...
↵ Enter to send · ⇧↵ New line

Questioning Page:



9.CONCLUSION

The AskDocs AI project successfully demonstrates the development of an intelligent document-based question-answering system that simplifies information retrieval from large and unstructured data sources. By integrating technologies such as Natural Language Processing, vector embeddings, and FAISS-based similarity search, the system enables users to upload documents and obtain accurate, context-aware answers through natural language queries. It overcomes the limitations of traditional keyword-based search by providing faster, more relevant, and user-friendly responses. Overall, the project achieves its objective of improving efficiency, reducing manual effort, and showcasing the practical application of AI in real-world document management systems.

10.FUTURE SCOPE

1. **Integration with Advanced AI Models**
   The system can be enhanced by integrating large language models (LLMs) to generate more detailed and conversational answers.
2. **Support for Multiple File Formats**
   Future versions can support additional formats such as Word documents, images (using OCR), and web content.
3. **Scalability Improvements**
   The system can be deployed on cloud platforms and optimized to handle large-scale data and multiple users efficiently.
4. **User Interface Development**
   A fully functional web or mobile application can be developed to improve user interaction and accessibility.
5. **Multilingual Support**
   The system can be extended to support multiple languages for both documents and user queries.
6. **Voice-Based Interaction**
   Voice input and output features can be added to allow users to interact with the system using speech.
7. **Advanced Features**
   Additional functionalities like document summarization, real-time collaboration, and personalized recommendations can be implemented

5. MILESTONE

S. No.	Project Activity	Estimated Start Date	Estimated End Date
1.	Project Allotment	29/01/2026	16/02/2026
2.	Synopsis Creation	18/02/2026	24/02/2026
3.	Synopsis Presentation	25/02/2026	26/02/2026
4.	Document Upload & PDF Parsing Module	27/02/2026	18/03/2026
5.	Embedding Generation & FAISS Integration	19/03/2026	29/03/2026
6.	Integration, Testing, and Bug Fixes	30/03/2026	19/04/2026
7.	Documentation	20/04/2026	25/04/2026
8.	Final Presentation	25/04/2026	30/04/2026

6. MEETING WITH THE SUPERVISOR

Date of the meet	Comments by the Supervisor	Signature of the Supervisor

7. REFERENCES

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[6] Pdfplumber. GitHub Repository, https://github.com/jsvine/pdfplumber
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[8] PostgreSQL Global Development Group. PostgreSQL Documentation, https://www.postgresql.org/docs